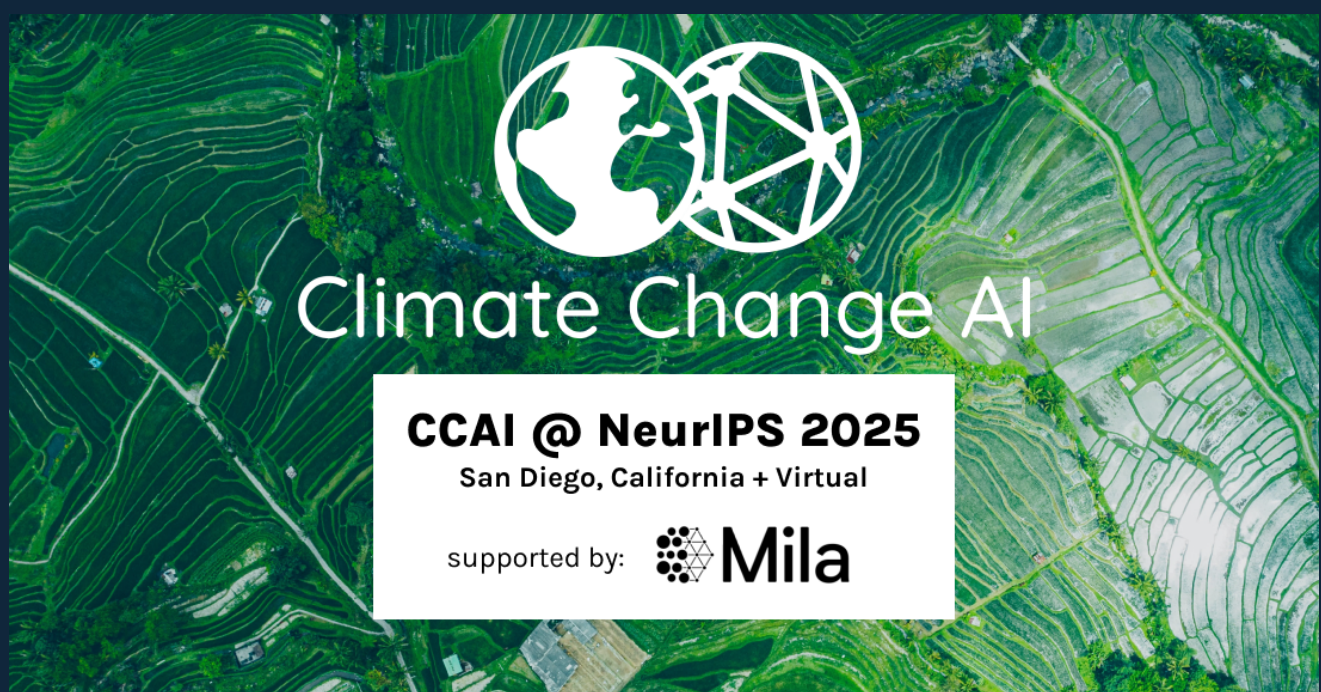
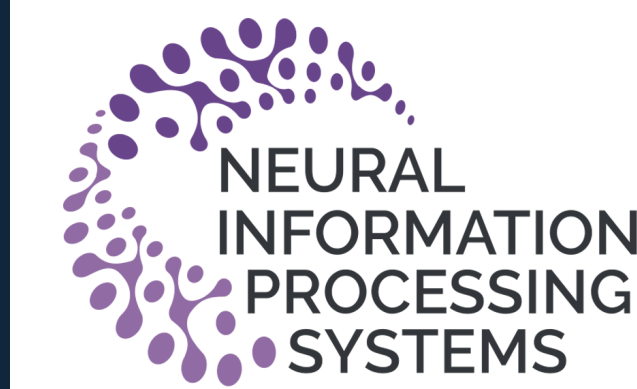




PiggyCast: IMPROVING WEATHER PREDICTION ACCURACY THROUGH A STACKING-BASED ENSEMBLE AI APPROACH



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Abstract

Recently, Artificial Intelligence Weather Prediction (AIWP) models have outperformed classical Numerical Weather Prediction (NWP) Models in various weather prediction benchmarking criteria. Given the paradigm shift from numerical to machine learning models, such forecasts can be generated in seconds to minutes on a standard laptop. In this study, we propose a supervised machine learning ensemble strategy, known as **PiggyCast** (a portmanteau of the words **piggyback** and **forecast**), trained on top of these forecast datasets (a method known as “stacking”) to predict ERA5 variables, thereby exploiting the strengths of each base model and aiming to outperform forecasts from any base model alone. The improvement in PiggyCast’s Root Mean Squared Error on Geopotential Height at 500 hPa, relative to the base models, was notable, with an increase in performance as forecast lead time increased.

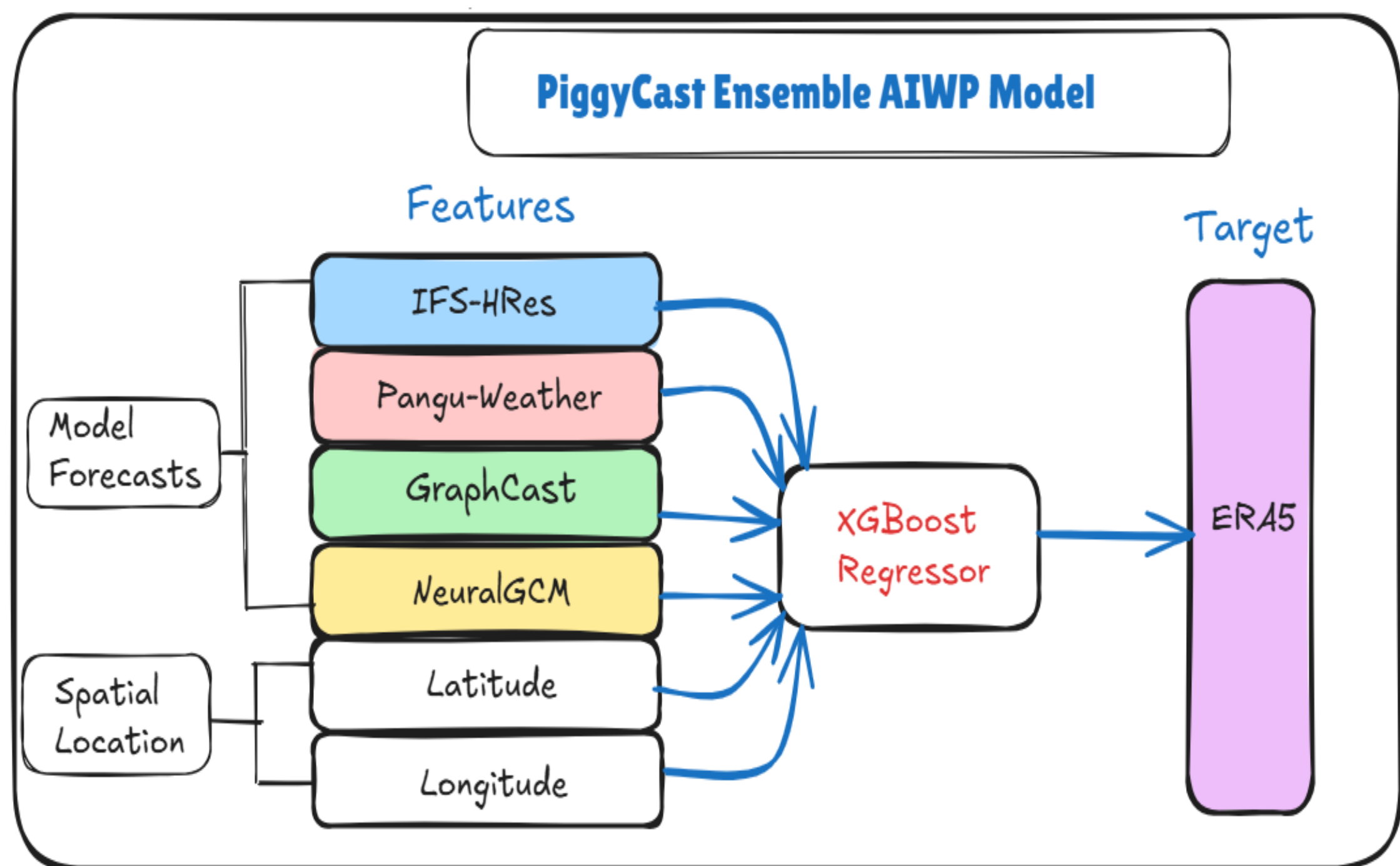


Figure 1. PiggyCast Model: Traditional machine learning ensemble model (stacking) trained on top of forecasts of current state-of-the-art weather prediction models. Image generated with Gemini.

Objectives

1. Develop and assess an **ensemble machine learning model** through **stacking** forecasts from the selected weather prediction models for **enhanced predictive performance**.
2. Investigate the **effect of input features** on the trained ensemble model for **interpretability and explainability** of the forecasting process.

Data Description

The data used in this study comprises the ERA5 reanalysis dataset and the forecasts of different AI, numerical and hybrid weather prediction models. The data are publicly accessible via the WeatherBench 2 framework [4], facilitating independent analysis, evaluation, and further research. We considered one atmospheric weather variable: **Geopotential Height at 500 hPa** for **9 lead times (48, 72, 96, 120, 144, 168, 192, 216 and 240 hours)** at longitude/latitude 64×32 (5.625°) spatial grid resolution and **12-hour** temporal resolution for the entire globe.

Table 1. Summary of various weather prediction models considered in this study

Model	Resolution	Forecast Range	Architecture
IFS HRES	0.1°	10 days	Traditional NWP
Pangu-Weather	0.25°	7 days - 10 days (scalable)	3D Vision Transformer
GraphCast	0.25°	10 days	Graph NN
NeuralGCM	$\sim 0.7^\circ$	10 days	Hybrid (GCM + ML)

Methodology

1. **PiggyCast with XGBoost Regressor(Objective 1):** XGBoost algorithm [1].

- Let $f_{t+\tau}^{(i)}(\mathbf{x})$ denote the forecasted value at spatial location $\mathbf{x}$ and lead time τ hours by model i . The feature vector for each instance is defined as:

$$\mathbf{x}_{\text{input}} = \left[f_{t+\tau}^{(\text{GraphCast})}(\mathbf{x}), f_{t+\tau}^{(\text{Pangu})}(\mathbf{x}), f_{t+\tau}^{(\text{NeuralGCM})}(\mathbf{x}), f_{t+\tau}^{(\text{IFS HRES})}(\mathbf{x}), \text{lon}(\mathbf{x}), \text{lat}(\mathbf{x}) \right], \quad (1)$$

with the target variable: $y = f_{t+\tau}^{(\text{ERA5})}(\mathbf{x})$.

- The loss function around $\hat{y}_t^{(t-1)}$ (prediction from the ensemble up to iteration $t-1$):

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_i(x_i) + \frac{1}{2} h_i f_i^2(x_i) \right] + \Omega(f_i), \quad (2)$$

where $g_i = \partial_{y^{(t-1)}} l(y_i, \hat{y}_i^{(t-1)})$ is the first-order gradient, and $h_i = \partial_{y^{(t-1)}}^2 l(y_i, \hat{y}_i^{(t-1)})$ is the second-order derivative (Hessian).

- Evaluation Metrics **Area-weighted RMSE**: Let M_1 and M_2 represent ERA5 value and a model spatial forecast, respectively, then

$$\text{RMSE}(M_1, M_2) = \sqrt{\frac{\sum_{i=1}^n w_i \cdot (M_{1,i} - M_{2,i})^2}{\sum_{i=1}^n w_i}}, \quad (3)$$

where $w_i = w(\phi_i) = \cos\left(\frac{\pi \cdot \phi_i}{180}\right)$, is the area weight for grid point i on latitude ϕ and n is the number of grid points.

Model Performance Across Lead Times: For each fold and model, we compute the RMSE, & average across folds:

$$\text{RMSE}_{\text{avg}}^{(m,\tau)} = \frac{1}{K} \sum_{k=1}^K \text{RMSE}_k^{(m,\tau)}, \quad (4)$$

where m is the model and $K = 10$ is the number of folds.

2. **Feature Attribution (Objective 2):** SHAP (SHapley Additive exPlanations) values [3].

- Given a trained model f and input $\mathbf{x}$, the SHAP value ϕ_j for feature j represents the marginal contribution of that feature:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^M \phi_j, \quad (5)$$

where ϕ_0 is the base value (mean model output) and M is the number of input features

Results

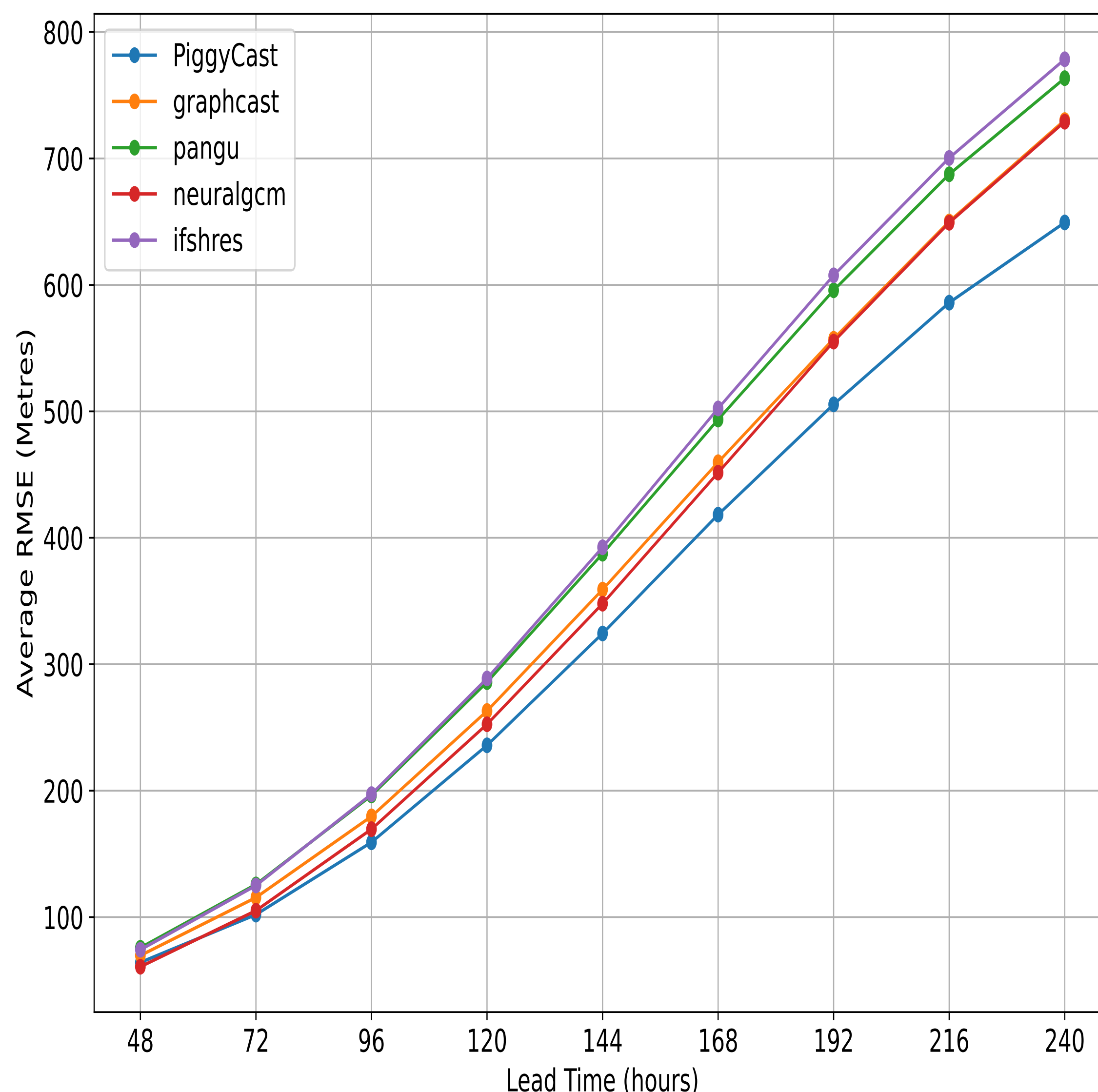


Figure 2. A plot of average area-weighted RMSE for all models, 48-240 hours lead times.

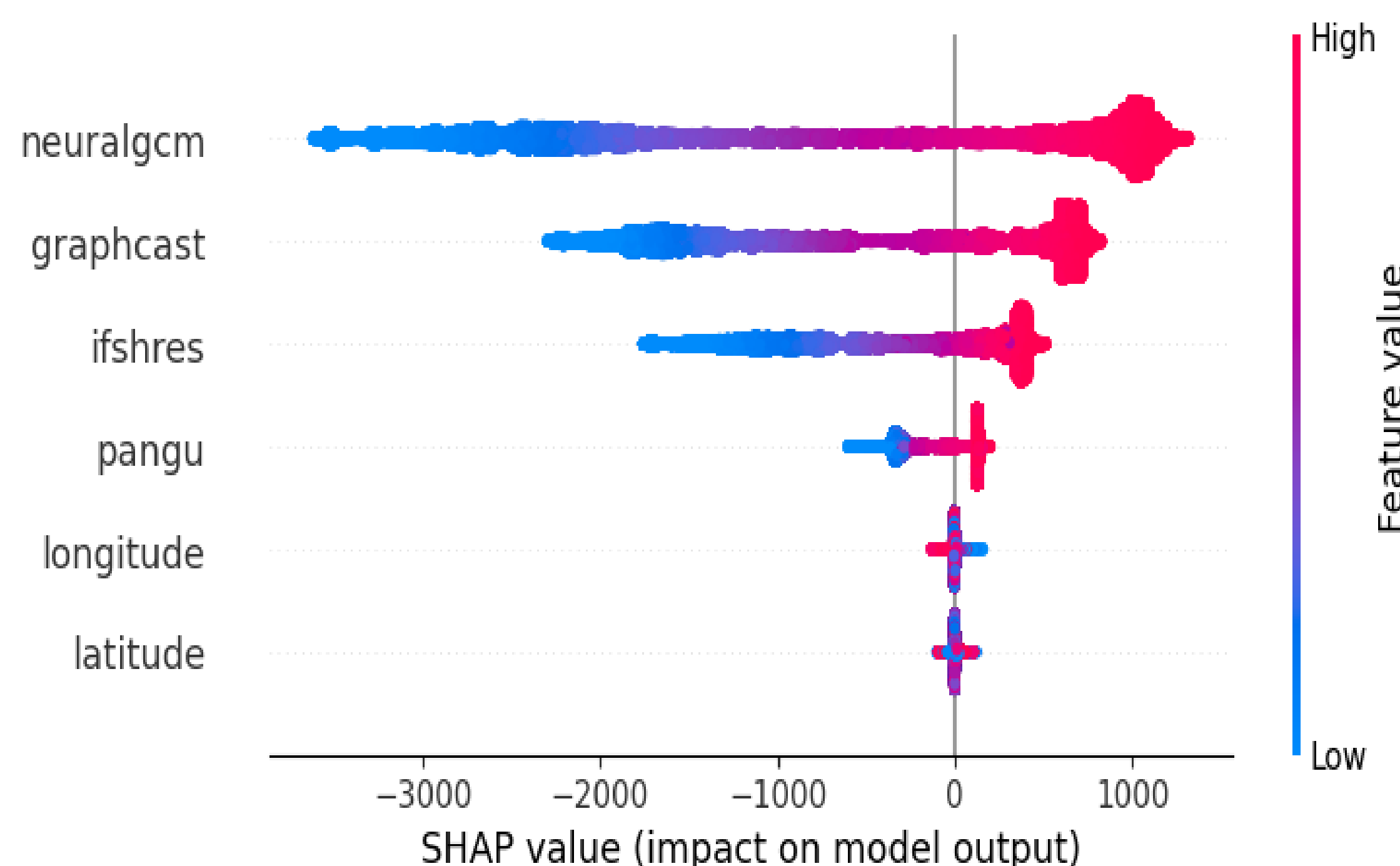


Figure 3. A beeswarm plot of SHAPley values of PiggyCast's features at 72-hour lead time.

Conclusion

Given the low compute cost of making forecasts, and that each frontier AIWP model has its strengths and limitations (depending on the weather variable, region of the globe, and forecast lead time), **we argue that the future of the most skilful weather forecasts will likely come from machine learning stacking**, by the very nature that stacking typically yields performance better than any base model alone.

Future Work

- Incorporation of **additional weather variables** and **weather prediction models**.
- **Testing the sensitivity of PiggyCast**, relative to the best base model, to the **spatial scale of a region**. Since in this study we optimise for weather globally, we suspect that when **targeting smaller spatial scales, PiggyCast may exhibit impressive results**.
- Extension of the framework to address **extreme weather events, nonstationary climate conditions** and **uncertainty quantification**.

References

- [1] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 785–794, 2016.
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